Lab Sheet 11

Live Motion & Security Alarm Detector (Mobile Camera + OpenCV)

Working

The camera feed is read frame by frame. Each frame is resized to 640x480, converted to grayscale and blurred. The previous frame is subtracted from the current one (absdiff), the result is thresholded and dilated, and contours are found. Contours larger than MIN_AREA are treated as motion: a green box is drawn, a red ALERT banner is shown, and a snapshot is saved to the snapshots/ folder at most once every 3 seconds.

config.py

```
# Phone camera (IP Webcam app) ka URL, apne phone ka IP daalo
CAMERA_URL = "http://10.124.189.186:8080/video"

# Laptop webcam se test karne ke liye upar wali line hata ke ye use karo:
# CAMERA_URL = 0
```

motion_detector.py

```
import cv2
import os
import time
from datetime import datetime
from config import CAMERA_URL

os.makedirs("snapshots", exist_ok=True)

MIN_AREA = 1500      # isse bada moving object = motion (kam karo to sensitive)
SNAPSHOT_GAP = 3     # seconds, itne gap par hi nayi photo save hogi

cap = cv2.VideoCapture(CAMERA_URL)
ret, frame = cap.read()

# Agar phone camera connect na ho to laptop webcam try karo
if not ret:
    print("Phone camera connect nahi hua, laptop webcam try kar raha hu...")
    cap = cv2.VideoCapture(0)
    ret, frame = cap.read()

if not ret:
    print("Camera connect nahi hua. IP aur Wi-Fi check karo.")
    exit()


def prepare(img):
    img = cv2.resize(img, (640, 480))
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    return img, cv2.GaussianBlur(gray, (21, 21), 0)


frame, prev_gray = prepare(frame)
last_saved = 0
photo_count = 0

while True:
    ret, frame = cap.read()
    if not ret:
        break

    frame, gray = prepare(frame)

    # 1. Frame subtraction
    diff = cv2.absdiff(prev_gray, gray)

    # 2. Threshold + dilate
    thresh = cv2.threshold(diff, 25, 255, cv2.THRESH_BINARY)[1]
```

```
    thresh = cv2.dilate(thresh, None, iterations=2)

    # 3. Contours
    contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL,
                                   cv2.CHAIN_APPROX_SIMPLE)

    motion = False
    for c in contours:
        if cv2.contourArea(c) < MIN_AREA:
            continue
        motion = True
        x, y, w, h = cv2.boundingRect(c)
        cv2.rectangle(frame, (x, y), (x + w, y + h), (0, 255, 0), 2)

    if motion:
        cv2.rectangle(frame, (0, 0), (640, 60), (0, 0, 255), -1)
        cv2.putText(frame, "ALERT: MOTION DETECTED!", (15, 40),
                    cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 255), 2)

        # snapshot save (har frame par nahi, gap ke saath)
        if time.time() - last_saved > SNAPSHOT_GAP:
            name = datetime.now().strftime("snapshots/alert_%Y%m%d_%H%M%S.jpg")
            cv2.imwrite(name, frame)
            print("Saved:", name)
            photo_count += 1
            last_saved = time.time()
    else:
        cv2.putText(frame, "Status: Clear", (15, 30),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.8, (0, 255, 0), 2)

    cv2.putText(frame, f"Photos saved: {photo_count}", (15, 470),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255, 255, 0), 2)

    prev_gray = gray

    cv2.imshow("Security Camera", frame)
    cv2.imshow("Threshold (debug)", thresh)

    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```

Output Screenshots

Snapshots saved automatically in the snapshots/ folder when motion was detected.

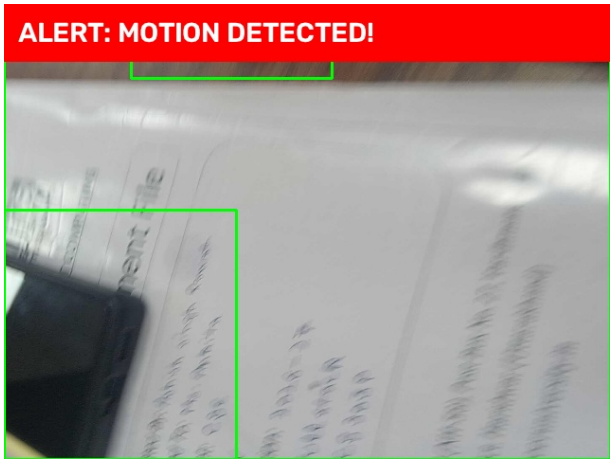


Fig 1: Lab file cover moved in front of the camera. Motion boxes appear on the moving phone edge and the top border of the document.

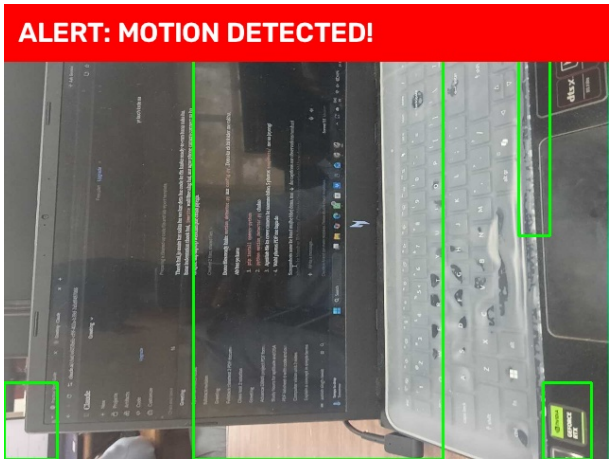


Fig 2: Camera pointed at a laptop and keyboard while moving. Large motion boxes cover the screen and keyboard areas.

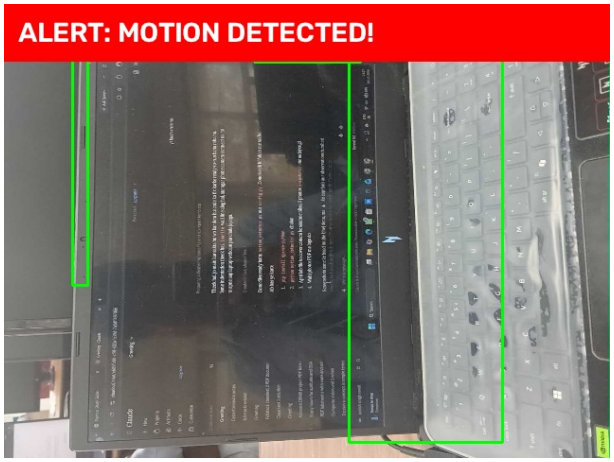


Fig 3: Laptop screen and keyboard in view. A motion box marks the laptop display and another marks the left edge.